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PROJECT SOFTWARE ARCHITECTURE

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1 Introduction & Goals

This document was prepared as part of the Software Architecture course at FCTUC-MEI, by students Artur Yurchuk, Leonardo Silva, Josemar Andrade and Tran Sy Khoa. The objective of the project is to design and document the architecture of a Wi-Fi Security System capable of monitoring, detecting, and mitigating network threats in home and small office networks. The document follows the arc42 template and includes a C4 model to support the architectural views.

1.1 Requirements Overview

This section presents the Functional Requirements that guide the design of the Wi-Fi Security System. These requirements capture the key functionalities that the system must provide to support network monitoring, threat mitigation, and device management. The table below summarizes each requirement, which is further addressed in the system's architecture, Use Cases, and Threat Mitigation strategies presented in this document.

Id	Functional Requirement	Explanation
1	Traffic Monitoring and Analysis	Continuously monitor network traffic to detect suspicious patterns and interception attempts.
2	Blocking Suspicious Devices and Unsafe Traffic	Automatically block devices suspected of intercepting data or using insecure protocols.
3	Encryption and Secure Communication	Ensure that all sensitive data is encrypted (e.g., WPA3) and support secure connections like VPN.
4	Bandwidth Management	Set bandwidth limits for devices or users to prevent one device from consuming the entire network.
5	Automatic Intrusion Detection	Monitor traffic to detect suspicious activities (e.g., brute force attacks) and automatically block malicious devices or IPs.
6	Strong Authentication for Camera Access	Implement strong authentication mechanisms (e.g., MFA) to protect access to security cameras.

Table 1: Summary of Functional Requirements

The identified Functional Requirements serve as the foundation for architectural decisions, ensuring that the system meets its security, usability, and maintainability objectives.

1.2 Driving Forces

The architecture of this system is influenced by several key driving forces that must be considered to meet both functional and non-functional requirements. These forces are fundamental for shaping design decisions, ensuring that the system is practical, scalable, secure, and adaptable. The main driving forces for the system architecture are:

- **Network Security:** Ensuring network security is critical, with a focus on real-time traffic monitoring, unauthorized access detection, and encryption
- **Automatic Threat Response:** The system must autonomously detect and block threats (e.g., unauthorized devices) without manual intervention. This ensures timely action against security breaches.
- **Hardware Limitations:** The system must run on low-resource hardware, like Raspberry Pi or old laptops, requiring optimized and efficient resource usage.

- **Scalability and Flexibility:** The system should be adaptable to various routers and ISPs, ensuring compatibility across different environments with minimal configuration.
- **Cost and Efficiency:** The solution must be cost-effective and use readily available hardware, avoiding the need for specialized, expensive components.

1.3 Quality Goals

This section defines the Quality Goals that the architecture of the Wi-Fi Security System aims to achieve. These goals provide a prioritized framework to guide design decisions, trade-offs, and system implementation. They ensure that the architecture addresses both functional and non-functional aspects such as security, usability, and maintainability, which are critical to the system's effectiveness and user acceptance. The following table summarizes these goals along with their respective priorities.

Priority	Quality Attribute	Quality Goal
1	Security	Ensure that the system detects and blocks unauthorized access, brute-force attacks, and data modification attempts quickly and effectively, with responses and alerts triggered as fast as possible.
2	Performance	Maintain overall network performance by limiting bandwidth usage of non-priority devices, ensuring that legitimate users do not experience slowdowns or interruptions, even under high load.
3	Availability	Ensure the network remains operational even during attacks or overloads, guaranteeing continuous access to essential services (such as cameras).
4	Scalability	Allow the system to remain functional and performant as the number of connected devices increases, by fairly allocating resources and detecting abnormal behavior.
5	Maintainability	Allow administrators to update configuration settings (e.g., bandwidth limits) without restarting the system, applying changes immediately.

Table 2: Consolidated Quality Goals by Attribute

These Quality Goals are systematically addressed throughout the architecture and are supported by the system's Use Cases, Components, and Threat Mitigation strategies.

1.4 Stakeholders

This section identifies the key stakeholders of the Wi-Fi Security System. Stakeholders are individuals or organizations who influence or are affected by the system's architecture and operation. Understanding their needs and expectations is essential to ensure that the system design aligns with its intended use and operational context. The following table summarizes the main stakeholders and their respective concerns.

Stakeholder	Description / Interest
End Users	Residents or workers in small offices who want to protect their Wi-Fi networks against unauthorized access through an affordable, effective, and easy-to-use solution.
Developers	Responsible for the design and technical implementation of the system, ensuring that functional and quality requirements are met.
Project Client	Defines the project's objectives, monitors its progress, and validates the results, especially in an academic project context.

Table 3: Stakeholders of the Wi-Fi Network Security System

2 Constraints

This section identifies and describes the main constraints that influence the architecture and design of the Wi-Fi Security System. Constraints are external or internal limitations that restrict design choices and architectural decisions. They can be technical (related to technology, tools, or standards) or business-related (derived from organizational, legal, or project factors). Addressing these constraints early in the design process helps ensure realistic and feasible architectural solutions.

2.1 Technical Constraints

This subsection describes the technical constraints that directly influence architectural decisions and design choices. These constraints arise from hardware limitations, compatibility requirements, and performance expectations. Addressing them ensures that the Wi-Fi Security System can operate effectively within the defined technical boundaries.

- **TC001 - Hardware Limitations:**

- Must run on low-cost hardware (Raspberry Pi or old laptops).
- Limited processing power and memory available.
- Cloud-based alternatives were considered but rejected due to latency and security concerns.

- **TC002 - Compatibility Requirements:**

- Must work with at least one router model or ISP.
- Future updates may expand router compatibility.
- External network monitoring appliances were considered but not chosen.

- **TC003 - Performance Constraints:**

- System must block unauthorized connections within 5 seconds (from QAS1.1).
- Bandwidth management changes must take effect immediately without restart (from QAS5).
- Must maintain network performance with increasing devices (from QAS4).

2.2 Business Constraints

This subsection identifies the business-related constraints that impact the architecture and implementation of the Wi-Fi Security System. These constraints arise from project budget, organizational resources, cost considerations, and legal and privacy requirements. They must be addressed to ensure that the system remains feasible, affordable, and aligned with its target market.

- **BC001 - Budget Limitations:**

- Must stay affordable for home/small office users.
- Use of open-source tools and existing hardware required.
- Minimal marketing and distribution costs (digital platforms preferred).

- **BC002 - Team Constraints:**

- Small, efficient team to keep costs low.
- Limited staffing resources.

- **BC003 - Cost of Ownership:**

- Initial purchase must be affordable (possibly freemium model).
- Must work with existing routers to avoid additional hardware costs.
- Minimal ongoing maintenance costs.

- **BC004 - Legal and Privacy Requirements:**

- The system must comply with applicable data privacy regulations (e.g., GDPR).
- User data must be protected and not shared with third parties.
- Logging must be done in accordance with privacy best practices.

3 Context and Scope

This section defines the context and scope of the Wi-Fi Security System. It describes how the system interacts with its environment, identifies external actors and systems, and clarifies the system boundary. Understanding this context helps position the architecture and provides a shared understanding of what is included in the system and what is external to it.

3.1 Business Context

The Wi-Fi Security System is intended to be deployed in home and small office networks to provide enhanced network security, device management, and threat mitigation. The primary users are Network Administrators and end users who wish to control and monitor their network. The system operates independently of ISP infrastructure and does not require modifications to ISP-provided router firmware. It complements existing network setups by acting as an intelligent gateway for security and monitoring.

The main business stakeholders are:

- **Network Administrators:** responsible for configuring and managing network security.
- **Network Users:** benefit from a more secure and controlled network environment.
- **ISP Providers:** indirectly impacted; the system is designed to be compatible with their infrastructure without requiring changes.

3.2 Technical Context

The Wi-Fi Security System interacts with several technical components:

- **ISP Router:** provides internet connectivity; configured to route traffic through the Wi-Fi Security System.
- **Admin Interface:** used by Network Administrators to configure and monitor the system; communicates with the backend via REST API.
- **Backend Services:** provide core system functionality such as traffic inspection, device management, and threat response.
- **API Client:** embedded in the Admin Interface; communicates securely with the Backend Services.
- **Connected Devices:** client devices on the LAN that send and receive traffic through the system.

All communication between components is secured via HTTPS where applicable. The system is designed to run on low-cost hardware within the local network environment.

3.3 Scope (In-scope vs. Out-of-scope)

This subsection defines the boundaries of the Wi-Fi Security System. It clarifies which functionalities and components are included within the scope of the system (In-scope) and which are explicitly excluded (Out-of-scope). This distinction ensures a shared understanding of the project limits and helps guide architectural decisions.

3.3.1 In-scope:

The following functionalities and capabilities are included within the scope of the system, as defined in the project scope and refined through architectural analysis:

- **Real-time monitoring of network activity:** The system will capture and analyze traffic in real time to detect anomalies and potential threats.
- **Detection of unauthorized devices:** The system will identify devices attempting to connect to the network without proper authorization.
- **Identification of security threats:** This includes detecting packet sniffing attempts, bandwidth abuse, and the presence of rogue access points.

- **Automated blocking of suspicious connections:** The system will be capable of autonomously blocking devices or traffic flows deemed suspicious or malicious.
- **User notifications:** The system will provide real-time alerts and notifications to Network Administrators regarding detected threats and actions taken.
- **Compatibility with at least one ISP/router brand:** The system will be tested and validated for compatibility with at least one supported ISP router model.
- **Operation on minimal hardware:** The system will run on low-cost hardware platforms such as Raspberry Pi or repurposed laptops, in line with project constraints.

3.3.2 Out-of-scope:

The following functionalities and components are explicitly excluded from the scope of the system. They are either not aligned with the project objectives, not feasible within current constraints, or considered out of priority for this implementation:

- **Cloud-based monitoring solutions:** The system is designed to operate locally without reliance on external cloud services, to reduce latency and improve privacy.
- **Support for all router brands/models initially:** The system will initially support only selected router models; full compatibility with all existing routers is outside the current scope.
- **Advanced hardware requirements:** The architecture is optimized for minimal hardware and will not require enterprise-grade servers or dedicated appliances.
- **Enterprise-grade security features:** The system targets home and small office environments and does not aim to provide enterprise-level security certifications or features.
- **Physical security components:** The system does not include integration with physical security hardware beyond network-connected cameras that use standard network protocols.
- **Mobile application development:** No dedicated mobile apps will be developed in this phase; administration will be performed via a responsive web interface.

3.4 System Interfaces

This subsection describes the main system interfaces of the Wi-Fi Security System. These interfaces define how the system communicates with its users, the underlying network infrastructure, and external components. A clear definition of interfaces helps ensure consistent integration and guides both implementation and testing.

- **SI001 - Administrator Interface (Web-based UI):**
 - Web-based management console accessible via standard web browser.
 - Device management: bandwidth configuration, device blocking/unblocking.
 - Real-time alert notifications and system status updates.
 - User authentication and access control.
- **SI002 - Network Interfaces:**
 - Communication with ISP Router (e.g., via API, SSH, or manual configuration).
 - In-line traffic monitoring and analysis (packet inspection at gateway level).
 - Bandwidth enforcement using traffic shaping and firewall rules.
- **SI003 - Device Interfaces (including Security Cameras):**
 - Network access control (allow/block per device, including security cameras and IoT devices).
 - Traffic monitoring for all connected devices.
 - Detection of abnormal behavior from connected devices (e.g., excessive bandwidth usage, unexpected communication patterns).

4 Solution Strategy

This section describes the fundamental solution strategy for the Wi-Fi Security System. It captures the key architectural decisions, technology choices, and organizational considerations that shape the design and implementation of the system. The goal is to ensure that the architecture aligns with the quality goals and is capable of addressing the identified threats while meeting the project's constraints and business context. The system is designed following a layered architecture approach, with modular components organized into monitoring, analysis, and enforcement layers. This modularity supports maintainability and scalability while ensuring a clear separation of concerns.

4.1 Motivation

The fundamental architectural and technological decisions aim to directly address the key threats identified in the project (data theft, network slowdowns, unauthorized access to IoT devices such as security cameras). In addition, the solution strategy is aligned with the defined quality goals (Section 1.3), ensuring that the system delivers high security, usability, and maintainability. The chosen solution is designed to operate efficiently on constrained hardware and within the budget and organizational constraints of the project.

4.2 Quality Goals and Solution Mapping

The following table maps the system's Quality Goals to specific threat scenarios and corresponding solution approaches. This mapping ensures that architectural and design decisions directly contribute to achieving the desired quality attributes while mitigating the identified threats.

Quality Goal	Scenario	Solution Approach
Security (QAS1)	Threat 01 – Theft of Personal/Financial Data	Real-time traffic monitoring + encryption (WPA3/VPN)
Performance (QAS2)	Threat 02 – Slower Internet	Bandwidth management + DDoS detection
Availability (QAS3)	Threat 03 – Unauthorized Camera Access	Multi-Factor Authentication (MFA) + firmware updates
Scalability (QAS4)	Increased connected devices	Fair bandwidth allocation + behavioral analysis
Maintainability (QAS5)	Admin updates bandwidth rules	Swift access to Web-based admin interface (no restart required)
Confidentiality (QAS6)	Unauthorized camera access	Encrypted video streams + strong authentication

Table 4: Mapping of Quality Goals to Threat Scenarios and Solution Approaches

4.3 Key Organizational Decisions

The project is subject to several organizational constraints and decisions that influence architectural and technology choices. The following table summarizes these key organizational decisions and their rationale.

Decision	Rationale
Use of Raspberry Pi	Meet hardware constraints (low cost, compatibility)
Open-Source Tools	Reduce development costs, leverage community support
Smaller team with predefined roles	More cost effective
Layered Architecture	Separates monitoring, analysis, and enforcement for modularity and easier maintenance

Table 5: Organizational Architecture Decisions

4.4 Technology Stack

The following table summarizes the selected technology stack for the Wi-Fi Security System. The chosen technologies and tools are aligned with the system's quality goals, project constraints, and target deployment environment (low-cost hardware in home and small office networks).

Decision Area	Technologies/Tools	Rationale
Monitoring	Snort, Wireshark	Lightweight, open-source network analysis
Authentication	OAuth 2.0, Multi-Factor Authentication	Secure access to cameras and admin interfaces
Database	PostgreSQL	ACID compliance for logging and audit trails
UI	Flask (Python)	Simple web interface for bandwidth management
Deployment	Docker	Containerization for scalability and portability; suitable for deployment on low-cost hardware (e.g., Raspberry Pi)

Table 6: Technology Stack Overview

5 C4 Model

This section presents the architectural views of the Wi-Fi Network Security System using the C4 model approach. The C4 model provides a hierarchical set of diagrams to visualize the system from a high-level context down to detailed component structures. This multi-level representation helps align the architecture with stakeholders' needs and supports both understanding and implementation.

The following subsections present the System Context Diagram, Container Diagram, and Component Diagrams.

5.1 System Context Diagram

The System Context Diagram provides a high-level overview of the Wi-Fi Network Security System, focusing on its interactions with external actors and systems. It establishes the boundaries of the system and identifies its primary inputs and outputs. This level is critical for understanding the external context in which the system operates, ensuring that the architecture aligns with the functional and non-functional requirements defined earlier.

The key elements around the main system (Wi-Fi Network Security System) of this diagram are:

- **Network Administrator:** The primary user of the system, responsible for managing network security, approving devices, and responding to alerts.
- **Unauthorized User (Attacker):** A potential threat trying to gain unauthorized access to the Wi-Fi network, posing the threats specified earlier.
- **ISP Router:** An external system that provides network connectivity and serves as the primary entry point for both legitimate users and potential attackers, and which is the primary container to be monitored and protected.

The main interactions are:

- The **Network Administrator** interacts with the system through a Web-based Admin Interface, configuring security settings, reviewing logs, and receiving alerts.
- The **ISP Router** communicates with the system to provide raw network traffic for analysis, while the system manages device access and control to maintain network security. For this it's necessary a configuration in the ISP Router to redirect all the network traffic, which is specified in section x.x.
- The **Unauthorized User (Attacker)** attempts to access the network illegally, potentially exploiting weaknesses in the router or network configuration.
- The **Wi-Fi Network Security System** has the ability to manage device access and the response to certain access of attacker or unauthorized devices.

This diagram represented visually at figure 1 establishes the overall context for the Wi-Fi Network Security System, laying the foundation for more detailed views at the container and component levels. The next sections will break down the internal structure and interactions within the system, providing a deeper understanding of its architecture.

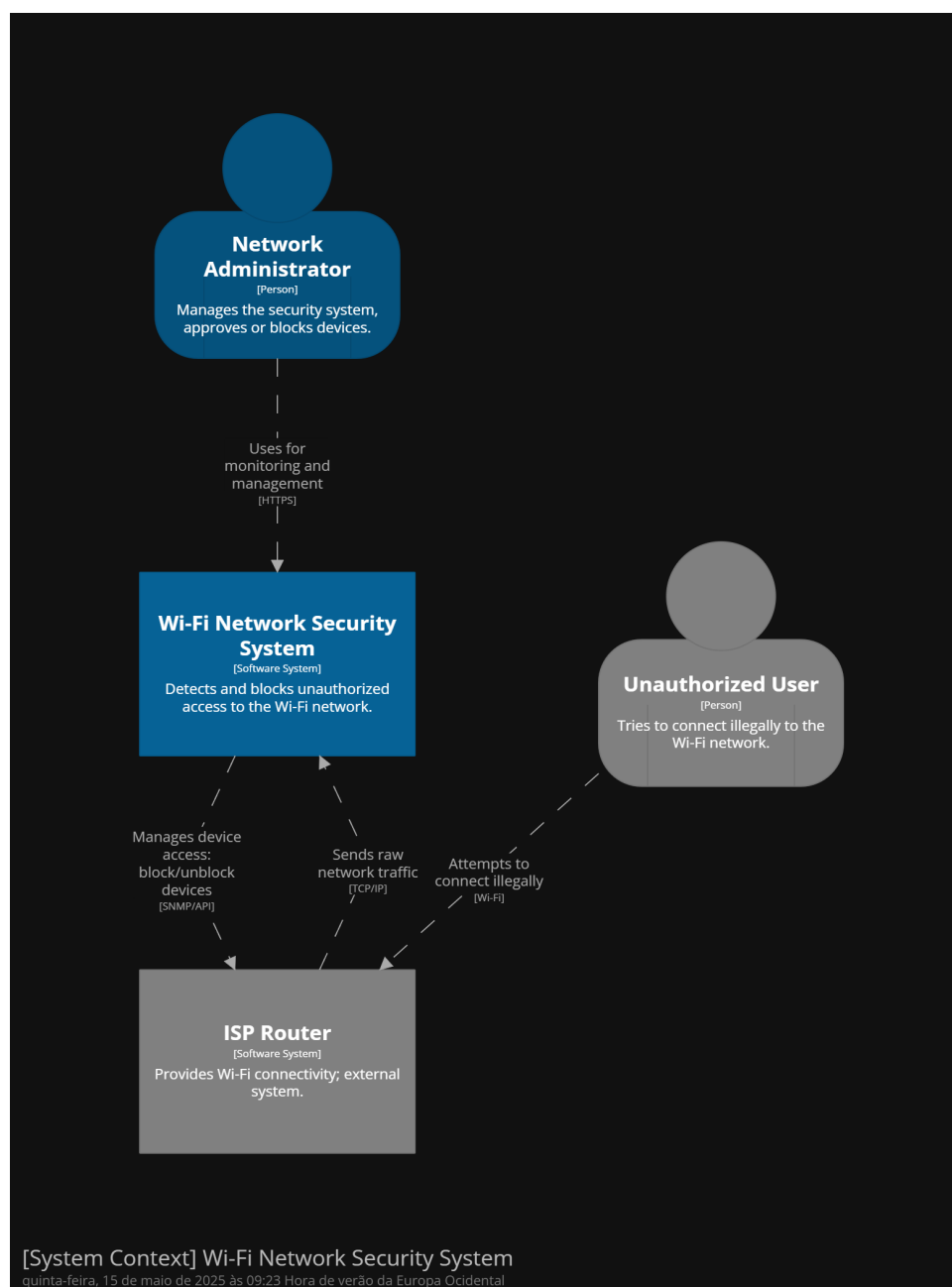


Figure 1: System Context Diagram for Level 1 of C4 Model

5.2 Container Diagram

The Container Diagram provides a more detailed view of the internal structure of the Wi-Fi Network Security System, breaking it down into its main containers and their interactions. This level focuses on the high-level technical architecture, including data flow, responsibilities, and the technologies used within each container.

The key components that represent this container are:

- **Traffic Encrypter:** Encrypts all incoming network traffic from the ISP Router before it is analyzed, ensuring data privacy and integrity.
- **Traffic Monitoring Service:** Captures and inspects network packets in real time, providing critical data for anomaly detection.
- **Threat Detection Engine:** Analyzes network traffic for suspicious patterns and behaviors, using rule-based or AI-driven algorithms to detect potential threats.

- **Admin Interface:** Provides a web-based dashboard for network administrators to monitor security events, review logs, and configure the security system.

The most important interactions between external and internal components are:

- The **ISP Router** forwards encrypted network traffic to the **Traffic Encrypter** due to a configuration made in ISP Router, which decrypts the data before passing it to the **Traffic Monitoring Service**.
- The **Device Management Module** communicates directly with the ISP Router to block or allow specific devices based on security policies.
- The **Network Administrator** interacts with the **Admin Interface** to manage devices, configure security settings, and review alerts.
- The **Traffic Monitoring Service** sends processed traffic data to the **Threat Detection Engine** for analysis.
- The **Threat Detection Engine** communicates with the **Device Management Module** to block potentially malicious devices.

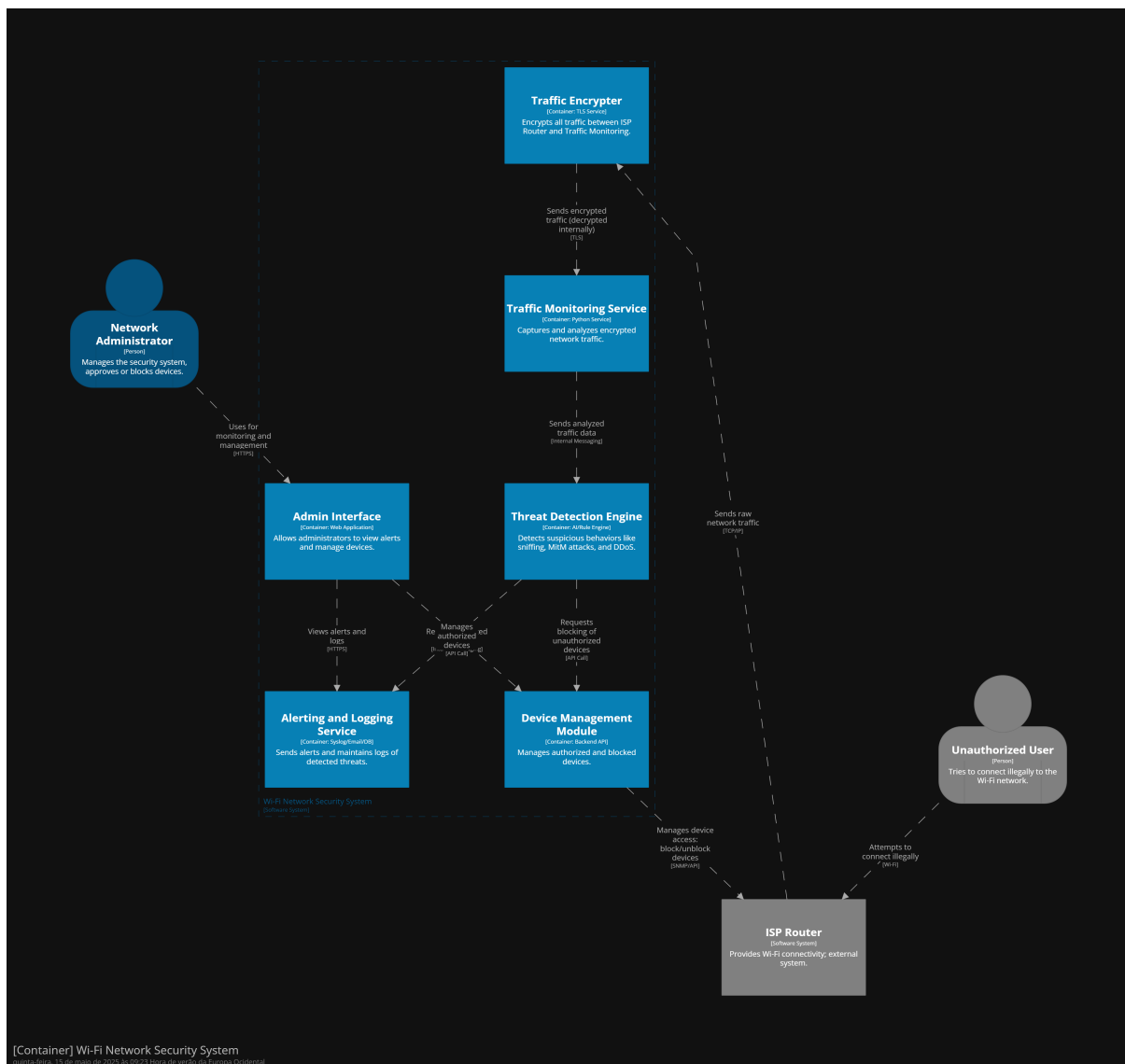


Figure 2: Container Wi-Fi Network Security System for Level 2 of C4 Model

5.3 Component Diagrams

The Component Diagram provides a detailed breakdown of the internal structure of key containers within the Wi-Fi Network Security System. It focuses on the individual components that make up each container, their responsibilities, and their interactions. This level is crucial for understanding the internal logic and data flows within the system, supporting both software development and system maintenance.

5.4 Admin Interface Component

This component is visually represented in figure 6 and there is an use case attached to this components which it's the UC05 - Manage Network Security.

- **Network Administrator:** Manages the network security system, approves or blocks devices, and interacts with the Admin Interface via a web browser.
- **Login & Auth UI:** Provides the login screen and handles user authentication with the backend by sending credentials and managing tokens for secure access.
- **Dashboard UI:** Displays threat alerts, network status, and system summaries to the administrator, pulling data from backend services via API Client.
- **Device Control Panel:** Enables the administrator to view all connected devices and perform management actions such as approving, blocking, or removing them.
- **API Client:** Facilitates communication between the UI components and backend services using HTTPS REST APIs, handling requests and responses.
- **Alerting and Logging Service:** Sends alerts and maintains logs of detected threats, storing events from the detection engine and administrator actions for review.
- **Device Management Module:** Manages authorized and blocked devices, providing API endpoints to update device states and enforce network-level rules.

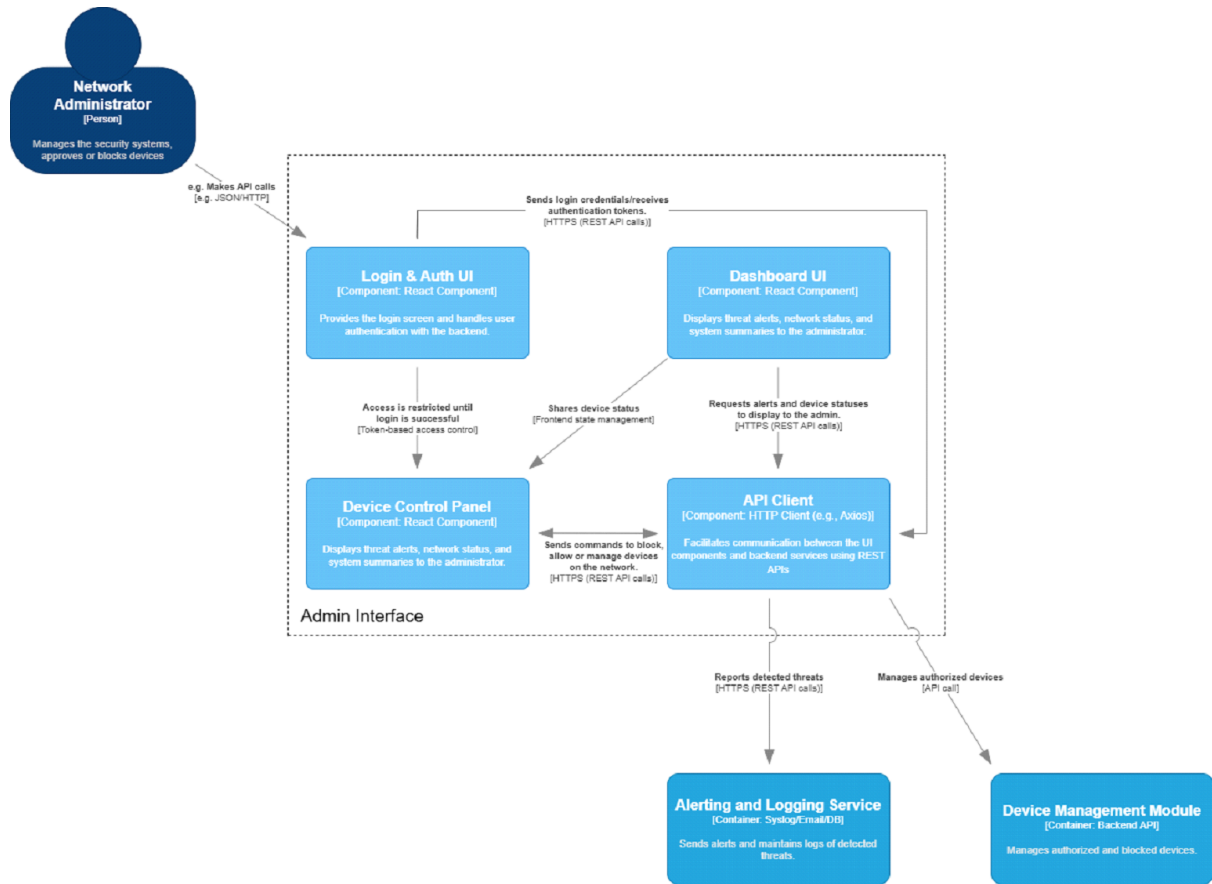


Figure 3: Component Admin Interface for Level 3 of C4 Model

Still about 6 the most important interactions between external and internal components are:

- The Network Administrator submits credentials through the Login & Auth UI, which communicates with the API Client to authenticate against backend services and receive tokens.
- The API Client sends device management commands from the Device Control Panel to the Device Management Module via REST API calls.
- The Device Management Module queries external Malware APIs to validate device threats during authorization check
- The Device Management Module reports detected threats to the Alerting and Logging Service, which stores logs in Synlog2EmailDB and triggers alerts.
- The Alerting and Logging Service pushes real-time threat notifications to the API Client, which updates the Device Control Panel for administrator visibility.
- The Admin Interface retrieves historical threat logs from the Alerting and Logging Service via the API Client for audit and analysis.
- The Device Control Panel fetches live network status and device authorization states from the Device Management Module through the API Client.

5.5 System's Components and Connectors

The following table provides a structured overview of the main architectural building blocks that make up the Wi-Fi Network Security System. It includes both:

- Components: active software modules responsible for specific functionality.

- Connectors: mechanisms that enable communication and coordination between components.

Each element is identified with a unique ID, along with its name, type, and a brief description of its role in the system.

This mapping is essential to understand how the system is modularized and how responsibilities are distributed across its parts. It also supports traceability between architectural decisions and the system's functional and quality requirements.

ID	Name	Type	Description
A01	Traffic Encrypter	Component	TLS service that decrypts incoming ISP traffic and re-encrypts it for internal processing.
A02	Traffic Monitoring Service	Component	Python service that captures packets, extracts metadata, and forwards it for analysis.
A03	Threat Detection Engine	Component	AI/Rule Engine applying signature and anomaly detection to identify threats.
A04	Device Management Module	Component	Backend API that maintains the device registry and pushes block/allow rules to the router.
A05	Alerting & Logging Service	Component	Syslog/Email/DB container that stores logs and dispatches alerts to administrators.
A06	Admin Interface	Component	React-based web app providing login, dashboards, device control, and real-time updates to the administrator.
A07	TLS	Connector	Encrypts all in-transit communication between containers and between UI and backend.
A08	HTTPS (REST API)	Connector	Standard request/response mechanism used by API Client and backend services for CRUD and management calls.
A09	Message Queue (e.g., RabbitMQ)	Connector	Decouples traffic monitoring, threat detection, and alerting for scalable, asynchronous event processing.
A10	WebSocket / SSE	Connector	Pushes real-time notifications and updates from backend services to the Admin Interface.
A11	Router Mirror Port	Connector	ISP router configuration that forwards a copy of all LAN traffic into the Traffic Encrypter for inspection.

Table 7: System Components and Connectors

6 Crosscutting Concepts

Crosscutting concepts are design principles or mechanisms that affect multiple parts of the system in a consistent way. In this project, we focus on crosscutting concepts related to safety and security—such as authentication, encryption, logging, error handling, and data retention—to ensure that every component of the Wi-Fi Network Security System handles sensitive data securely and protects the network from unauthorized access and threats.

6.1 Summary of Restrictions / Requirements

Business Constraints (BC)

1. **BC001 - Budget Limitations:** Controlling and minimising expenses related to system operations.
2. **BC002 - Team Constraints:** Guaranteeing that the system can be developed and maintained by a small team.
3. **BC003 - Cost of Ownership:** Guaranteeing that the system has low initial and maintenance costs.
4. **BC004 - Legal and Privacy Requirements:** Guaranteeing compliance with applicable data privacy regulations.

Technical Constraints (TC)

1. **TC001 - Hardware Limitations:** Handling hardware component restrictions and ensuring efficient operation on low-cost hardware.
2. **TC002 - Compatibility Requirements:** Guaranteeing compatibility with at least one supported router model or ISP, with support for future expansion.
3. **TC003 - Performance Constraints:** Ensuring that performance requirements are met regarding blocking unauthorized connections and managing bandwidth.

Use Cases (UC)

1. **UC01:** Block Device After Suspicious Behavior is Detected.
2. **UC02:** Set Bandwidth Limits per Device.
3. **UC03:** Configure and Monitor Network Security Policies.
4. **UC04:** Block Flagged Device.
5. **UC05:** Configure ISP Router.
6. **UC06:** Deploy and Configure the Security System.
7. **UC07:** Manage Authorized Devices.
8. **UC08:** Push Configuration to ISP Router.
9. **UC09:** Configure API Client with REST API Endpoints.

6.2 Solutions Presented

6.2.1 BC001 - Budget Limitations

- Monitoring tools: Leverages open-source network monitoring tools (Snort, Suricata) for traffic analysis.
- Admin Interface (A06): Built with free, open-source React to eliminate UI licensing costs.
- Device Management Module (A04): Uses Python/Flask (OSS) for all API logic with no license fee.
- Alerting & Logging Service (A05): Utilizes free database engines (SQLite/PostgreSQL) and syslog.

6.2.2 BC002 - Team Constraints

- Reusable API design: Implements consistent and stateless REST API patterns across components.
- Modular Components (A01–A06): Small, vertical-slice services (Encrypter, Monitor, Detector, etc.) allow a small team to own and maintain each piece.
- Shared Connectors (A07–A10): Common use of TLS, REST, queues, and WebSockets reduces duplicated implementation effort.

6.2.3 BC003 - Cost of Ownership

- Stateless components: Uses stateless service architecture for backend modules where possible.
- TLS 1.3 (A07) & HTTPS (A08): Uses standard protocols to avoid proprietary solutions.
- Message Queue (A09): Open-source brokers (RabbitMQ/Redis) used for event handling with minimal operational overhead.
- Router Mirror Port (A11): Uses existing router features to avoid additional capture hardware.

6.2.4 BC004 - Legal and Privacy Requirements

- The system complies with applicable data privacy regulations such as GDPR.
- All sensitive data is encrypted in transit using TLS.
- Audit trails are maintained for key administrative actions.

6.2.5 TC001 - Hardware Limitations

- Headless system: Runs the system in headless mode without a graphical user interface.
- Router Mirror Port (A11): Offloads packet capture to the router, reducing CPU/memory load on the host.
- Traffic Encrypter (A01): Lightweight TLS library optimized for ARM-based devices.
- Traffic Monitoring Service (A02): Built on efficient Python libraries (Scapy, PyShark), tuned for low-power hardware.

6.2.6 TC002 - Compatibility Requirements

- Plugin-based architecture: Designs the Router Communicator as a plugin-based component to support multiple router APIs.
- Standard Router API Adapter (A10): Implemented to support OpenWRT and common vendor APIs.
- Protocol-Agnostic Device Registry (A04): Accepts devices by MAC, IP, or hostname for broad router compatibility.
- Admin Interface (A06): Reads router and network settings from a central config store (A07), supporting multiple network topologies and ISP environments.

6.2.7 TC003 - Performance Constraints

- CPU affinity tuning: Applies CPU affinity to assign critical traffic monitoring and detection processes to specific CPU cores.
- Asynchronous I/O: Utilizes asynchronous I/O patterns in traffic processing and event streaming components.
- In-Memory Caching: Caches authorized-device lists and configuration in Redis or in-process LRU to speed up lookups.

- Real-Time Processing Pipeline: Chains packet sniffer → metadata extractor → rule engine with minimal queuing overhead to meet sub-5 second block latency.
- Dynamic Thresholds: Allows administrators to adjust detection sensitivity and bandwidth limits at runtime.

6.2.8 UC01 - Block Device After Suspicious Behavior is Detected

- UC01 is supported through real-time traffic monitoring (A02), threat detection (A03), and automated blocking via Device Management Module (A04).

6.2.9 UC02 - Set Bandwidth Limits per Device

- UC02 is supported via bandwidth configuration in the Admin Interface (A06) and enforcement through the Router Communicator (A10).

6.2.10 UC03 - Configure and Monitor Network Security Policies

- UC03 is implemented through the Admin Interface (A06), which provides policy configuration and monitoring dashboards.

6.2.11 UC04 - Block Flagged Device

- UC04 is supported by allowing the administrator to manually block flagged devices via the Admin Interface (A06) and Device Management Module (A04).

6.2.12 UC05 - Configure ISP Router

- UC05 involves setting up traffic redirection through Router Mirror Port (A11) and router configuration through the Admin Interface (A06).

6.2.13 UC06 - Deploy and Configure the Security System

- UC06 is supported through the system's installation process and configuration via Admin Interface (A06).

6.2.14 UC07 - Manage Authorized Devices

- UC07 is implemented in the Admin Interface (A06) with backend support from Device Management Module (A04).

6.2.15 UC08 - Push Configuration to ISP Router

- UC08 is supported via the Router Communicator module (A10), enabling remote configuration of ISP routers.

6.2.16 UC09 - Configure API Client with REST API Endpoints

- UC09 is implemented through the use of REST APIs (A08) secured via TLS (A07), enabling the Admin Interface (A06) to interact with backend services.

7 Architectural Decisions

This section presents the key architectural decisions made during the design of the Wi-Fi Security System. These decisions involve significant trade-offs and long-term impact on the system's availability, security, and ease of use. Each decision is based on the system's constraints, quality goals, and functional requirements, as well as the limited hardware and budget context defined in the project scope.

7.1 Deployment as Inline Gateway (vs Passive Sniffer or External Appliance)

Decision: Deployed the Wi-Fi Security System inline, acting as the default gateway for all LAN devices, enabling real-time monitoring, blocking, and traffic control.

Rationale: Inline deployment allows the system to actively enforce policies (block, throttle, or drop traffic), not just monitor. Passive sniffing is easier to deploy but offers no control over traffic.

Impact:

- Enables full control over network behavior
- Supports advanced features like real-time blocking and enforcement
- Makes the system effective in Security, Availability, and Performance, by detecting and mitigating threats instantly

7.2 Admin Interface as a Web-Based Local-Only Dashboard

Decision: Implement a local-only web interface to configure and monitor the system.

Rationale: A local dashboard provides easy access from any browser on the LAN, without exposing sensitive controls to the internet.

Impact:

- Preserves user privacy and minimizes exposure
- Makes the system compatible with most home/office environments
- Supports Security, Usability, and Maintainability with a simple, device-independent admin experience

7.3 Internal Communication via RESTful API (vs Message Queues or Shared DB)

Decision: Use internal RESTful APIs over HTTPS for communication between components.

Rationale: REST is simple to implement, test, and extend. It provides clean separation and is compatible with most platforms.

Impact:

- Reduces coupling between components
- Facilitates debugging, extension, and component reuse
- Enhances Maintainability, Security (with HTTPS), and Scalability

8 Quality Requirements

The Quality Requirements for this system have been thoroughly defined and justified throughout this document. This section provides a brief summary and references the relevant sections where these requirements are detailed.

- **Section 1.3 – Quality Goals:** Introduces the main Quality Attributes that guide the architecture, including Security, Performance, Availability, Scalability, Maintainability, and Confidentiality.
- **Section 4.2 – Quality Goals and Solution Mapping:** Maps the Quality Goals to identified threat scenarios, ensuring that each threat is addressed through specific architectural mechanisms.
- **Section 16 – Quality Attributes:** Provides a detailed description of each Quality Attribute and explains how the system architecture supports and enforces them.

In summary, the architecture of the Wi-Fi Network Security System has been carefully designed to meet its Quality Requirements, ensuring a solution that is secure, performant, maintainable, and scalable.

9 Risks & Technical Debt

This section presents the identified risks and unresolved issues that may impact the **Wi-Fi Network Security System** project. It also highlights key dependencies that the success of the project relies upon.

9.1 Unresolved Issues

- Integration testing with multiple ISP router models is still pending.
- Full performance testing under high device load is not yet completed.
- GDPR compliance audit is scheduled but not yet executed.
- Automatic firmware update handling for routers has not been fully validated.

9.2 Known Risks

- **Hardware Limitations:** Low-cost devices (Raspberry Pi) may face performance limitations with high device counts.
- **Router API Changes:** ISP firmware updates may break router API compatibility.
- **Security Threat Evolution:** New attack types may bypass current detection mechanisms.
- **Latency under Load:** Maintaining sub-5 second blocking performance with high traffic is challenging.
- **Small Team Size:** Limited team resources may delay development and maintenance.
- **Open Source Dependencies:** Risks if tools like Suricata, RabbitMQ, Flask become deprecated.
- **User Misconfiguration:** Incorrect configuration by admins could reduce system effectiveness.

9.3 Known Dependencies

- Integration with router APIs (OpenWRT, common vendor APIs).
- Regulatory compliance with GDPR and network monitoring laws.
- Availability of required hardware (Raspberry Pi, routers).
- Community support for open-source tools (Suricata, Snort, Redis, RabbitMQ).
- Admin training to ensure correct system use.

10 Threats

10.1 Threat 1 - Theft of Personal and Financial Data

One of the major security risks in an unsecured or compromised network is traffic interception. The perpetrator using packet sniffing techniques can capture sensitive information such as login credentials and personal data which can lead to identity theft, unauthorized account access, or financial fraud.

Cybercriminals often exploit this vulnerability which poses severe risks, including financial loss, identity theft, and unauthorized transactions.

10.2 Threat 2 - Slower internet

When someone who shouldn't have access gets into your Wi-Fi, they can use a lot of internets, making it slow or even making the network unusable for everyone else. Hackers can also spy on your online activity, steal your info, or even crash your network. This can happen if you share your password, have weak security, or don't monitor who's connected.

10.3 Threat 3 - Unauthorized Access to Security Cameras

Unauthorized access to network-connected security cameras poses serious risks, including privacy breaches, surveillance manipulation, and network infiltration. Attackers often exploit weak passwords, outdated firmware, and unencrypted data transmissions to gain control over these devices. Once compromised, cameras can expose sensitive footage, be disabled to bypass security monitoring, or serve as entry points for further cyberattacks.

Common scenarios include hackers accessing cameras with default credentials, exploiting firmware vulnerabilities, or using a compromised camera to infiltrate other devices on the network. Implementing strong authentication, encryption, and regular firmware updates is essential to mitigating these risks.

11 Misuse Cases

A misuse case is a diagram used to identify and describe situations where the system is used improperly or maliciously, typically with the intent to exploit flaws or vulnerabilities. It helps anticipate potential risks and weaknesses in the system's design, allowing for improvements in security and others.

In the next topics are detailed the misuse cases identified for each threat described previously. The first 3 topics refer to the threat 1, "*Theft of Personal and Financial Data*". The misuse cases numbered from 4 to 6 are referred to the threat number 2, "*Slower internet*". The misuse case numbered from 7 to 9 are refer to the threat 3, "*Unauthorized Access to Security Cameras*"

11.1 Misuse Case 1 (Threat 01) - Network Sniffing

- **Actors:** Malicious attacker, user.
- **Description:** An attacker gains access to an Wi-Fi network and uses packet-sniffing tools to intercept unencrypted sensitive information such as login credentials and credit card numbers.
- **How it happens:** Attackers gain access to the private network and after that use freely available packet-sniffing tools to capture unencrypted network traffic and in consequence can obtain important private information.
- **Impact:** Theft of personal information such as financial data or login credentials of important personal matters
- **How to prevent:** The system continuously monitors network activity in real-time (FR01) and identifies patterns that indicate suspicious behavior, such as packet sniffing. When such activity is detected, it automatically blocks the device responsible (FR02) and sends an alert to the administrator. To ensure that intercepted data cannot be read, all sensitive traffic on the network is encrypted using strong protocols like WPA3 or VPN (FR03). This combination of detection, encryption, and automated blocking prevents attackers from accessing or exploiting unencrypted personal or financial information.

This approach contributes directly to the Security quality attribute by ensuring that unauthorized access and data interception attempts are stopped quickly and effectively. It also supports Availability, as the system reacts autonomously to attacks without affecting the operation of the network.

11.2 Misuse Case 2 (Threat 01 & Threat 03) - Man-in-the-Middle (MitM) Attack

- **Actors:** Malicious attacker, victim user, online service.
- **Description:** The attacker intercepts communications as an man-in-the-middle between the victim and a legitimate website or in the case of Threat 03, positions himself between the security camera and the monitoring server , modifying or stealing transmitted data.
- **How it happens:** Attackers set up a rogue access point or exploit ARP spoofing techniques to intercept communication between the user and a legitimate service. In sequence the attacker can manipulate the data in real time, redirecting users to fraudulent websites or stealing sensitive information.
- **Impact:** Attackers can alter banking transactions, steal passwords, or inject malicious content.
- **How to prevent:** To prevent interception and manipulation of communications, the system enforces strong encryption protocols such as WPA3 or VPN (FR03), ensuring that any intercepted traffic remains unreadable. When the system detects suspicious communication behavior it automatically blocks the responsible device (FR01, FR02). In the case of access to security cameras (Threat 03), multi-factor authentication (FR06) ensures that even if communication is intercepted, attackers cannot authenticate or take control of the device. These measures strongly reinforce Security, by ensuring confidentiality and integrity of transmitted data, and contribute to Availability, by protecting the network from communication disruption or hijacking.

11.3 Misuse Case 3 (Threat 01) - Exploiting Weak Router Security

- **Actors:** Malicious attacker, network.
- **Description:** Attackers exploit outdated router firmware, weak passwords, or default credentials to take control of the router
- **How it happens:** Attackers take advantage of routers with outdated firmware, default credentials, or with clear vulnerabilities. Once access is gained, they can change DNS settings, monitor network traffic, or deploy malware onto connected devices.
- **How to prevent:** The system actively monitors device behavior to detect common exploitation patterns, such as unauthorized login attempts or suspicious changes in traffic flow. When such behavior is identified the system enters in the process of blocking the device involved (FR02). Even if default credentials or outdated firmware are exploited, behavioral analysis helps flag the intrusion early. This response ensures robust Security, preventing attackers from taking control of the router or connected devices, and enhances Scalability, as new threats are handled without requiring manual reconfiguration.

11.4 Misuse Case 4 (Threat 02) - Unauthorized Users Slowing Down the Internet

- **Actors:** Neighbor, Former guest or Hacker.
- **Description:** An unauthorized user connects to your Wi-Fi and consumes excessive amounts of data by streaming videos, downloading large files, or running data-heavy applications
- **How it happens:** Someone connects to Wi-Fi without permission (e.g., weak password or shared login), and they use a lot of data (e.g., streaming, downloading, or other heavy usage)
- **Impact:** The internet slows down and the legitimate users experience lag or have trouble using it properly
- **How to prevent:** The system enforces strict per-device bandwidth limits (FR04), which ensures that no single device can monopolize the available internet connection. If an unknown or unauthorized device is detected using excessive bandwidth, it is automatically flagged and blocked (FR02). The administrator can also adjust limits in real time, preventing service degradation for legitimate users. This mechanism upholds Performance, by ensuring fair usage across the network, and reinforces Security, by removing unauthorized users that could interfere with the network's quality of service.

11.5 Misuse Case 5 (Threat 02) - Hacker Overloading the Internet (DDoS Attack)

- **Actors:** Hacker.
- **Description:** DDoS (Distributed Denial of Service) attack is when a hacker floods a network or website with too much traffic, causing it to slow down or crash.
- **How it happens:** The hacker gains access to the network (The attacker may exploit a flaw in owner's router or use a stolen password), and uses the network to send a massive amount of data. They may also use your network to continuously download or upload large files.
- **Impact:** This overwhelms the router, causing slow speeds or crashes, making it impossible to use, which leads to legitimate users can't access the network
- **How to prevent:** The system continuously analyzes traffic and applies intrusion detection techniques to identify excessive or abnormal data flows (FR05). When a device generates an unusually high number of requests — a sign of DDoS — the system immediately blocks that device and sends an alert to the administrator (FR02). This rapid response ensures that the network remains stable and functional. The approach protects Availability, by preventing service disruption, and supports Security, by reacting effectively to network abuse or hostile behavior.

11.6 Misuse Case 6 (Threat 02) - Hidden devices Consuming the Internet

- **Actors:** Hacker or malicious user.
- **Description:** An unauthorized device connects to the Wi-Fi network and consumes data without the owner's knowledge.
- **How it happens:** The hacker gains access to the network (The attacker may exploit a flaw in owner's router or use a stolen password), and sneaks a divide into the Wi-Fi network. They hide their device so the owner doesn't see it.
- **Impact:** The internet gets slower, and the owner might not notice until the internet gets to that state and he can't find out the reason.
- **How to prevent:** The system maintains a real-time inventory of all devices connected to the network and continuously tracks their bandwidth usage (FR04). Unauthorized devices, even if not visible to the user, are detected through traffic fingerprinting and flagged. Once identified, these hidden devices are blocked automatically (FR02), and alerts are issued. This proactive approach ensures that network Performance is not degraded by hidden data consumption, while enhancing Security by exposing and removing hidden threats from the network.

11.7 Misuse Case 7 (Threat 03)- Unauthorized Access via Default or Weak Credentials

- **Actors:** Malicious attacker, security camera, legitimate user.
- **Description:** The attacker attempts to access the security camera using default or weak credentials that were not changed by the user after installation.
- **How it happens:** The attacker scans for vulnerable security cameras still using default credentials provided by the manufacturer. Once found, they can access the video feed and manipulate the device's settings.
- **Impact:** The attacker can view live footage, compromise the privacy of the monitored area, alter camera settings, or even disable the camera, compromising physical security.
- **How to prevent:** The system requires the use of strong, complex passwords for any device with sensitive access, such as security cameras (FR06). Additionally, multi-factor authentication is enforced, significantly increasing the difficulty of unauthorized access. If repeated login attempts are detected, the system blocks the device or user account and notifies the administrator. These layers of protection directly support Security, by preventing unauthorized access, and ensure Confidentiality, by safeguarding private data streams and video feeds.

11.8 Misuse Case 8 (Threat 03) - Exploiting Firmware Vulnerability

- **Actors:** Malicious attacker, security camera, legitimate user.
- **Description:** The attacker exploits a vulnerability in the camera's firmware to gain remote control of the device.
- **How it happens:** The attacker searches for known vulnerabilities in the camera model and uses an exploit to bypass authentication and access the camera. After gaining control, the attacker can view footage, alter settings, or even deploy malware.
- **Impact:** The attacker can gain unrestricted access to the camera, compromise privacy, manipulate the footage, or use the camera as an entry point for other attacks on the network.
- **How to prevent:** The system detects abnormal device behavior that may indicate exploitation of firmware vulnerabilities, such as unusual port access, unexpected firmware requests, or changes in device behavior (FR05). When such anomalies are detected, the device is immediately isolated from the network. Moreover, secure communication (FR03) prevents data from being exfiltrated,

even if the device is compromised. These defenses contribute to Security, by limiting the impact of firmware-related attacks, and support Availability, by ensuring that essential services like camera streams remain uninterrupted.

12 Functional Requirements

12.1 FR01 (Threat 01) – Traffic Monitoring and Analysis for Interception Detection

In order to prevent threats numbered as 1, network traffic interception and data theft, the system must continuously monitor network traffic to detect suspicious sniffing patterns and potential interception attempts.

Acceptance Criteria:

- The system must analyze network traffic in real-time and identify suspicious packets.
- The system must generate alerts whenever sniffing patterns are detected on the network.
- The system must allow the administrator to view detailed logs of suspicious events.

12.2 FR02 (Threat 01) - Blocking of Suspicious Devices and Unsafe Traffic

Referring to the threat numbered as 1, the system must block devices suspected of intercepting traffic or using insecure protocols, and unauthorized devices as well. Acceptance Criteria:

- The system must automatically block connections from suspicious devices.
- The system must allow the user to approve or unblock devices manually.
- The system must support the configuration of custom rules to define safe and unsafe traffic.
- The system must block unauthorized devices connected to the internet.

12.3 FR03 (Threat 01) - Traffic Encryption and Secure Communication

Referring to the threat numbered as 1, the system must encrypt the traffic in order to prevent malicious attackers from intercepting sensitive data. Acceptance Criteria:

- Ensure all sensitive data transmitted over the network is encrypted using WPA3.
- Provide automatic alerts when a device attempts to downgrade encryption protocols.
- Support VPN tunneling for high-risk transactions.

12.4 FR04 (Threat 02) – Bandwidth Management

Limits bandwidth usage per device or user. Allows you to set bandwidth limits for individual devices or users to prevent one device from consuming all the available internet.

Acceptance Criteria:

- The router must allow the administrator to set bandwidth limits for individual devices or users (e.g., 5 Mbps per device).
- The router must provide a real-time bandwidth usage dashboard to monitor and adjust limits
- The system must enforce bandwidth limits without requiring a restart or manual intervention.
- The router must notify the administrator if a device exceeds its bandwidth limit.

12.5 FR05 (Threat 02) – Automatic Intrusion Detection: Detects and blocks suspicious activity.

Malicious activity refers to intentional actions like hacking, data theft, or spreading malware that harms a system or network, and it can be detected by monitoring unusual behavior, known threats, or security breaches. Detection can be achieved by:

- **Traffic monitoring:** The router analyzes incoming and outgoing traffic for patterns that match known threats (e.g., brute force attacks, port scanning).

- **Behavioral Analysis:** The system learns what normal traffic looks like and flags deviations (e.g., a device suddenly sending large amounts of data).
- Automatically blocks suspicious devices or IP addresses. Alerts are sent to the network administrator for further investigation.

Acceptance Criteria:

- The router must monitor network traffic for signs of malicious activity (e.g., brute force attacks, port scanning).
- The router must use behavioral analysis to detect unusual activity (e.g., a device suddenly sending large amounts of data).
- The router must send alerts to the administrator when suspicious activity is detected or blocked.
- The system must provide a log of all detected threats and actions taken.

12.6 FR06 (Threat 03) – Strong Authentication for Camera Access

In order to prevent unauthorized access to security cameras, the system must enforce strong authentication mechanisms to ensure only authorized users can access and manage the cameras.

Acceptance Criteria:

- The system must require strong passwords for all users accessing the cameras (minimum of 12 characters, including uppercase, lowercase, numbers, and symbols).
- The system must support multi-factor authentication (MFA) for users when logging into the camera system.
- The system must block user accounts after a set number of failed login attempts (e.g., 5 attempts).

13 Use Cases

In Software Development, understanding user requirements and effectively translating them into functional features is crucial for success.

A use case is a definition of a specific business objective that the system needs to accomplish. It describes the interaction between a system and its users or external entities.

Some system functionalities, such as:

- Real-Time Traffic Monitoring
- Automatic Detection of Sniffing / Interception Patterns
- Traffic Encryption (WPA3 / VPN)
- Behavioral Analysis for Anomaly Detection
- Failed Login Attempt Tracker (for Cameras)

And many others are implemented as services that operate without direct user interaction. While they are critical for security in each one of the Functional Requirements, they are not represented as explicit Use Cases, since Use Cases focus on interactions between users and the system. These features are still essential and are fully detailed in the architecture and functional requirements sections.

13.1 UC01 - Block Device After Suspicious Behavior is Detected

Use Case ID	UC01 - Block Device After Suspicious Behavior is Detected
Actors	Wi-Fi Network Security System, Network Administrator, Admin Interface
Use Case Name	Block Device After Suspicious Behavior is Detected
Description	Enables the system to block devices identified as malicious, suspicious, or unauthorized, based on real-time traffic monitoring and analysis. The administrator can also manually block devices that have been flagged as suspicious. Supports mitigation of threats such as DDoS, ARP Spoofing, and other unauthorized activities.
Preconditions	• The Wi-Fi Network Security System has detected abnormal traffic through the ISP Router. • The administrator may also have identified suspicious activity via the Admin Interface. • Blocking rules are enabled (automatic or manual approval mode).
Main Flow	1. The system identifies a device as suspicious through traffic analysis. 2. The system checks the blocking policy configuration for the detected suspicious traffic. 3. If auto-blocking is enabled, the system blocks the device immediately. 4. The blocked device is listed in the dashboard for review.
Alternate Flows	• 3a. If auto-blocking is not enabled, the system flags the device but waits for admin confirmation before blocking. • 3b. The administrator logs in and approves or rejects the action. This approval process is further detailed in UC04.
Postconditions	• The suspicious device is blocked (or pending approval). • A log is created with details of the event for auditing purposes.

Table 8: Use Case UC01 - Block Device After Suspicious Behavior is Detected

13.2 UC02 - Set Bandwidth Limits per Device

Use Case ID	UC02 - Set Bandwidth Limits per Device
Actors	Network Administrator, Admin Interface
Use Case Name	Set Bandwidth Limits per Device
Description	Allows the Network Administrator to configure and apply bandwidth limits to specific devices on the network, ensuring fair usage and helping mitigate threats such as excessive bandwidth consumption (Threat 02).
Preconditions	• The Network Administrator is authenticated and has access to the Admin Interface. • Devices are already connected to the network and recognized by the system.
Main Flow	1. The Network Administrator accesses the Admin Interface via a web browser. 2. The system displays a list of all connected devices. 3. The Network Administrator selects a device from the list. 4. The Network Administrator defines the desired bandwidth limit and confirms the configuration. 5. The system updates the configuration and applies the bandwidth limit in real time via the network traffic controller, without requiring a restart.
Alternate Flows	• If the selected device disconnects from the network, the system stores the configured bandwidth rule and automatically applies it when the device reconnects.
Postconditions	• The bandwidth limit is applied to the specified device. • The configuration is stored persistently in the system database. • The new limit is visible in the Network Administrator dashboard and enforced by the traffic controller.

Table 9: Use Case UC02 - Set Bandwidth Limits per Device

13.3 UC03 - Configure and Monitor Network Security Policies

Use Case ID	UC03 - Configure and Monitor Network Security Policies
Actors	Network Administrator, Admin Interface
Use Case Name	Configure and Monitor Network Security Policies
Description	Allows the Network Administrator to monitor the overall security status of the network, review threat alerts, and configure security policies such as auto-blocking thresholds, bandwidth protection rules, and device authorization criteria.
Preconditions	• The Network Administrator has valid credentials. • The system is actively monitoring network traffic. • The system and ISP router are available.
Main Flow	1. The Network Administrator logs into the Admin Interface via a web browser. 2. The system authenticates the administrator and displays the security dashboard. 3. The Network Administrator reviews the current security status and threat alerts. 4. The Network Administrator accesses the Security Policies configuration section. 5. The Network Administrator adjusts the desired policies (e.g., auto-blocking thresholds, DDoS protection levels, bandwidth protection rules). 6. The system validates and applies the updated security policies. 7. The action is logged, and the administrator receives a confirmation.
Alternate Flows	• 6a. If policy validation fails (e.g., conflicting thresholds), the system notifies the administrator and prevents applying invalid settings. • 2a. If there is no Network Administrator with the provided credentials, an error message is displayed and access is denied.
Postconditions	• The updated security policies are applied to the system. • A log entry is created with event details. • The security dashboard reflects the updated policies.

Table 10: Use Case UC03 - Configure and Monitor Network Security Policies

13.4 UC04 - Block Flagged Device

Use Case ID	UC04 - Block Flagged Device
Actors	Network Administrator, Admin Interface, Wi-Fi Network Security System
Use Case Name	Block Flagged Device
Description	Allows the Network Administrator to block devices that have been flagged as suspicious by the Wi-Fi Network Security System. This action is performed via the Admin Interface to prevent unauthorized or malicious devices from accessing the network.
Preconditions	• The Wi-Fi Network Security System has detected abnormal traffic through the ISP Router, flagged the suspicious device, and notified the Network Administrator.
Main Flow	1. The Network Administrator logs into the Admin Interface via a web browser. 2. The Admin Interface displays all flagged suspicious devices. 3. The Network Administrator selects and blocks the suspicious device. 4. The blocked device is listed in the dashboard for review.
Alternate Flows	• 3a. If the Network Administrator chooses not to block the device, the device is unflagged and regains full access to the internet.
Postconditions	• The suspicious device is either blocked or unflagged. • A log is created with details of the event for auditing purposes.

Table 11: Use Case UC04 - Block Flagged Device

13.5 UC05 - Configure ISP Router

Use Case ID	UC05 - Configure ISP Router
Actors	Network Administrator, Admin Interface, ISP Router
Use Case Name	Configure ISP Router to Route Traffic Through the Security System
Description	Describes the one-time configuration required on the ISP-provided router to ensure that all network traffic is routed through the Wi-Fi Security System for monitoring, policy enforcement, and threat mitigation.
Preconditions	• The ISP Router is already installed and working as the default gateway for the local network. • The Wi-Fi Security System is installed and properly connected to the local network.
Main Flow	1. The Network Administrator accesses the ISP Router configuration interface. 2. The Network Administrator configures the ISP Router to route LAN traffic through the Wi-Fi Security System's IP (as the default gateway or via static routing). 3. The Network Administrator configures additional routing settings and/or port forwarding rules as required to ensure that traffic is correctly redirected through the Wi-Fi Security System. 4. The configuration is saved, and the router reboots.
Alternate Flows	• 2a. If the ISP Router does not support advanced configuration, the Wi-Fi Security System may be connected in bridge mode between the ISP Router and internal devices.
Postconditions	• All network traffic from connected devices passes through the Wi-Fi Security System before reaching the internet. • The Wi-Fi Security System is now able to monitor, detect, and respond to network misuse scenarios and security threats effectively.

Table 12: Use Case UC05 - Configure ISP Router

13.6 UC06 - Deploy and Configure the Security System

Use Case ID	UC06 - Deploy and Configure the Security System
Actors	Network Administrator, Admin Interface, Wi-Fi Security System
Use Case Name	Initial Setup and Activation of the Wi-Fi Security System
Description	Describes the installation and configuration steps of the Wi-Fi Security System to receive and process network traffic, acting as a gateway for LAN devices. Ensures that the system is ready to enforce security policies and mitigate potential network threats.
Preconditions	• The ISP Router has been configured to forward traffic to the system (see UC05). • The system hardware (e.g., Raspberry Pi, laptop) is powered and connected to the local network. • The Network Administrator has access to the system's operating system and Admin Interface.
Main Flow	1. The Network Administrator installs and boots the Wi-Fi Security System. 2. The Network Administrator configures the system's network interface with a static IP address. 3. The ISP Router has already been configured to forward LAN traffic (see UC05), and the system is set as the default gateway for all LAN devices. 4. IP forwarding is enabled on the system to allow routing between interfaces. 5. Firewall and packet inspection tools are initialized. 6. The system starts monitoring and enforcing security policies. 7. The Network Administrator verifies that connected devices have internet access and that traffic is being monitored via the Admin Interface and system logs.
Alternate Flows	• 5a. If inspection tools fail to start, the system falls back to monitoring-only mode, without active enforcement of security policies.
Postconditions	• The Wi-Fi Security System is fully operational and able to inspect, block, and report on network activity. • The system is functioning as the primary gateway or inspection point for the local network and is enforcing security policies.

Table 13: Use Case UC06 - Deploy and Configure the Security System

13.7 UC07 - Manage Authorized Devices

Use Case ID	UC07 - Manage Authorized Devices
Actors	Network Administrator, Admin Interface, Device Management Module
Use Case Name	Manage Authorized Devices
Description	Allows the Network Administrator to view, add, remove, or modify the list of devices authorized to access the network. This functionality ensures that only approved devices are granted access, enhancing overall network security. This functionality contributes to the mitigation of threats related to unauthorized device access (Threat 02), and supports network security and access control policies.
Preconditions	• The Network Administrator is authenticated in the Admin Interface. • The Device Management Module is operational. • There are connected devices or historical device connection data available.
Main Flow	1. The Network Administrator accesses the Admin Interface via a web browser. 2. The Network Administrator navigates to the “Authorized Devices” section. 3. The Admin Interface displays the current list of authorized devices. 4. The Network Administrator can: • Add a new device (e.g., by MAC address) • Remove a device from the list • Edit a device’s permissions or name 5. The system applies the changes and updates the access control rules in real time.
Alternate Flows	• 5a. If the device is already in the list, the system notifies the Network Administrator and prevents duplication. • 5b. If the Network Administrator tries to remove a currently connected device, the system prompts for confirmation.
Postconditions	• Changes are enforced immediately on the network. • The authorized device list is updated. • A log entry is created detailing the performed action for auditing purposes.

Table 14: Use Case UC07 - Manage Authorized Devices

13.8 UC08 - Push Configuration to ISP Router

Use Case ID	UC08 - Push Configuration to ISP Router
Actors	Network Administrator, Wi-Fi Security System, Admin Interface, ISP Router
Use Case Name	Push Configuration to ISP Router
Description	Allows the Wi-Fi Security System to remotely apply configuration changes to the ISP Router (such as routing adjustments, firewall rules, or device block-lists). This enables centralized management, enforces consistent security policies, and improves maintainability. The process is initiated by the administrator through the system's Admin Interface.
Preconditions	• The ISP Router supports remote configuration via API, SSH, or similar protocol. • The Wi-Fi Security System has administrative credentials or appropriate access permissions. • Network connectivity between the system and the ISP Router is available.
Main Flow	1. The Network Administrator accesses the Admin Interface of the Wi-Fi Security System. 2. The administrator initiates a configuration update. 3. The system validates its access to the ISP Router. 4. The system prepares the configuration changes. 5. The system sends the changes to the ISP Router using the available interface. 6. The system verifies if the changes have been applied successfully. 7. A success message is displayed to the administrator.
Alternate Flows	• 3a. If access to the ISP Router fails, the system notifies the administrator and aborts the operation. • 5a. If the ISP Router does not support automated updates, the system provides manual configuration instructions. • 6a. If verification fails, the system retries the operation and logs the error.
Postconditions	• The ISP Router is updated with the new configuration. • The operation is logged for audit purposes. • The network operates under the updated security and routing policies.

Table 15: Use Case UC08 - Push Configuration to ISP Router

13.9 UC09 - Configure API Client with REST API Endpoints for requests in Admin Interface

Use Case ID	UC09 - Configure API Client with REST API Endpoints for requests in Admin Interface
Actors	Network Administrator, Admin Interface, API Client, Wi-Fi Security System Backend Services
Use Case Name	Configure API Client with REST API Endpoints for requests in Admin Interface
Description	Allows the API Client in the Admin Interface to be configured with REST API endpoints to perform secure requests to the Wi-Fi Security System backend services. Enables real-time interaction with system functionalities such as device management, bandwidth control, and threat alerts. Supports usability (QAS2) by providing a responsive admin interface, and contributes to security (QAS1) through secure API communication.
Preconditions	• The Admin Interface is running and accessible. • The backend services expose REST API endpoints secured via HTTPS. • The API Client is initialized and has valid access credentials (tokens) if required.
Main Flow	1. The Admin Interface initializes the API Client on startup. 2. The API Client is configured with the list of REST API endpoints (URLs, methods, authentication tokens). 3. The Admin Interface triggers API requests (e.g., get device list, block device, configure bandwidth). 4. The API Client performs the request using HTTPS. 5. The API Client receives and processes the response. 6. The Admin Interface updates the UI with the response data.
Alternate Flows	• 4a. If the API endpoint is unavailable or the request fails, the API Client retries, logs the failure, and/or displays an error to the Network Administrator. • 2a. If authentication tokens are invalid or expired, the API Client initiates a secure token refresh or re-authentication flow.
Postconditions	• The API Client is correctly configured and can perform secure REST API calls. • The Admin Interface is able to interact with the Wi-Fi Security System backend in real time. • Actions performed via the API are securely logged in the system for auditing purposes.

Table 16: Use Case UC09 - Configure API Client with REST API Endpoints for requests in Admin Interface

14 Quality Attributes

Quality attributes define essential characteristics of the system that influence its behavior, performance, and security. Quality attributes are the most difficult architectural drivers to identify, define, and test. Quality attributes are so important that they condition architecture much more than functional requirements.

14.1 QAS1 - Security

Security is the main quality attribute that we can point out to in this work. Security *"refers to the degree to which a software system safeguards the information or data so that users or other systems have the degree of access to these data based on the authorization level"*.

It relates to threat 1,2 and 3 and so we took different possible scenarios for each threat

For threat number 1 quality we have (QAS1.1):

- **Source** - Unauthorized device
- **Stimulus** - The system detects and denies an unauthorized connection attempt.
- **Artifact** - Network access control system.
- **Environment** - Occurs during the normal operation
- **Response** - The system blocks the unauthorized connection within 5 seconds and logs the attempt.
- **Response Measure** - The system successfully prevents unauthorized access in more than 95% of cases, and alerts are sent within 2 seconds.

For threat number 2 we can have (QAS1.2):

- **Source** - Hacker
- **Stimulus** - A hacker attempts to brute-force the Wi-Fi password.
- **Artifact** - The router's intrusion detection and prevention system.
- **Environment** - During an active attack
- **Response** - The router detects the brute force attempt, blocks the hacker's IP address, and alerts the administrator.
- **Response Measure** - The hacker is unable to gain access, and the network remains secure.

Associated with the Security for threat number 1 and 3, we must also consider **Integrity**, although both are related, integrity focuses on the issue in which data transactions are maintained (QAS1.3):

- **Source** - Malicious Attacker
- **Stimulus** - The attacker intercepts and modifies network packets.
- **Artifact** - Secure communication protocol.
- **Environment** - Secure transaction mode enabled.
- **Response** - The system detects and blocks any attempts that can compromise any data.
- **Response Measure** - No unauthorized data modifications detected in more than 95% of monitored transmissions.

14.2 QAS2 - Performance

Related to the scenario of Network Bandwidth Management mentioned in the Functional Requirements section, Performance is the main quality attribute that we want to focus and improve, *"performance is the most critical software quality attribute as it can make the user system to a hung state"*

- **Source** - Unauthorized user
- **Stimulus** - A user starts streaming 4K videos on the network.
- **Artifact** - The router's bandwidth management feature.
- **Environment** - Normal operating conditions.
- **Response** - The router limits the guest user's bandwidth to a fraction of the total available bandwidth.
- **Response Measure** - The size of the fraction varies by the amount of the actual user. For example, each person has equal amount of bandwidth, but more users also mean bandwidth for each gets smaller. The Network has a certain limit of users. The main network's performance remains unaffected, with no noticeable slowdown for legitimate users

14.3 QAS3 - Availability

Related to the scenario of Network Overload, Availability is the main quality attribute, availability is directly related to Reliability. Availability is *"the percentage of time the application is available for use"*, and in a more general way we can say that reliability and availability are *"the ability of software applications to behave as expected and function under the maximum possible load."* For threat number 2 quality we have (QAS3.1):

- **Source** - A device
- **Stimulus** - Multiple devices are connected to the network, and one device starts consuming excessive bandwidth
- **Artifact** - The router's bandwidth management and intrusion detection feature,
- **Environment** - During peak usage time.
- **Response** - The router limits the bandwidth for the offending device and ensures fair usage for all connected devices.
- **Response Measure** - The network ensures users can connect without interruptions, and the network stays fast, with no lag or delays, even if one device is using a lot of bandwidth.

For **Threat 3** (Unauthorized Access to Security Cameras), we can define the following scenario (QAS3.2):

- **Source:** External attacker
- **Stimulus:** The attacker attempts to overload the network by repeatedly trying to access the security cameras without authorization, causing a denial of service (DoS).
- **Artifact:** Camera bandwidth management and access control system.
- **Environment:** During peak usage time, when multiple cameras and devices are connected to the network.
- **Response:** The system detects and blocks the attack, limiting the bandwidth consumption of the affected camera while keeping other cameras and devices operating without interruption.
- **Response Measure:** The network continues to operate normally, without loss of connectivity to the cameras or significant performance degradation, maintaining availability in over 95% of cases.

14.4 QAS4 - Scalability

When there is an increased number of devices connected to the network, Scalability is the most important quality attribute to focus on because, Scalability *"is how easily your system can handle increasing demands without affecting the application's performance"*

Scalability and Availability are in some ways related, because they have the same goal in this case, that is to provide good network access to all the devices and actors.

- **Source** - Devices
- **Stimulus** - The number of connected devices increases significantly.
- **Artifact** - The router's bandwidth and device management feature.
- **Environment** - During normal operation.
- **Response** - The router efficiently manages the increased load by allocating bandwidth fairly and monitoring for suspicious activity.
- **Response Measure** - The network performance remains stable, with no degradation in speed or reliability

14.5 QAS5 - Maintainability

Maintainability *"refers to how easily software developers can add new features and update existing features with new technologies"*

In this project Maintainability is important in the following scenario:

- **Source** - Network Administrator
- **Stimulus** - The administrator needs to change the bandwidth limits for connected devices.
- **Artifact** - The router's bandwidth management interface.
- **Environment** - During normal operation.
- **Response** - The administrator can access the bandwidth management settings and adjust the limits .
- **Response Measure** - Accessing the bandwidth management settings must be swift (no more than a minute). Also, the changes take effect immediately, with no need to restart the router.

14.5.1 QAS6 - Confidentiality

Confidentiality is the main quality attribute related to protecting sensitive data, ensuring that only authorized users can access video footage from security cameras. Confidentiality refers to "the protection of data from unauthorized access and disclosure."

For **Threat 3** (Unauthorized Access to Security Cameras), we can define the following scenario (QAS6.1):

- **Source** - External attacker
- **Stimulus** - The attacker attempts to access live or recorded video footage from the cameras by exploiting weak passwords or default credentials.
- **Artifact** - The security camera's authentication and encryption mechanisms.
- **Environment** - Cameras are operating on a network with outdated security settings or without encryption.
- **Response** - The system denies access to unauthorized users and encrypts all transmitted data, ensuring that the video footage remains private.
- **Response Measure** - Unauthorized access attempts are blocked in more than 95% of cases, and no sensitive footage is exposed.

14.6 Constraints

14.6.1 Technical

Technical constraints are fixed technical design decisions that absolutely cannot be changed. Most often technical constraints are provided by stakeholders (perhaps after some digging) at the outset of the project. Sometimes a team may choose to create a constraint, perhaps to simplify the world in which they are working. In this project, the main Technical Constraints that we considered are

Hardware Limitations: The system must run on low-cost hardware, such as a Raspberry Pi or an old laptop, with limited processing power and memory.

Compatibility with Routers: The system must be compatible with at least one router model or ISP to ensure it can monitor and block unauthorized access, and the whole architecture should work easily. Future updates could expand support to more router brands. Also alternatively using external network monitoring appliances instead of relying on router compatibility.

14.6.2 Business

Business constraints are unchangeable business decisions that in some way restrict the software architecture design. Business constraints are similar to technical constraints in that they are decisions that cannot be changed, but rather than influencing structures directly through technology, the influence occurs indirectly through business decisions. In this project, the main Business Constraints that we considered are:

Budget The project must stay within a limited budget to remain affordable for home and small office users. By our choice the development costs should be optimized by using open-source tools and existing hardware where possible. Marketing and distribution expenses should be minimal, possibly leveraging digital platforms.

Team composition and make-up Minimal Staffing Requirement – The team should remain small and efficient to keep costs low.

Cost of Ownership There is various costs to take into account, they are:

- **Initial Purchase Cost** – has to be affordable, possibly a one-time purchase or a free model with premium features.
- **Hardware Costs** – has to work with existing routers or low-cost devices to avoid additional expenses for users.

15 Architecture

15.1 C4 Model

This section presents the architecture of the system following the C4 model approach, which includes three abstraction levels: the system context, the container view, and the component view. Each level provides a different perspective of the system, allowing stakeholders to understand how the solution is structured, interacts with external actors, and fulfills the functional and non-functional requirements defined earlier.

15.1.1 Level 1 : System Context Diagram

The System Context Diagram provides a high-level overview of the entire system within the environment. It shows the key external actors (administrators, routers, and attackers) and how they interact with the security system. This level clarifies the system's boundaries, external dependencies, and primary interfaces.

This diagram it's presented in figure 1.

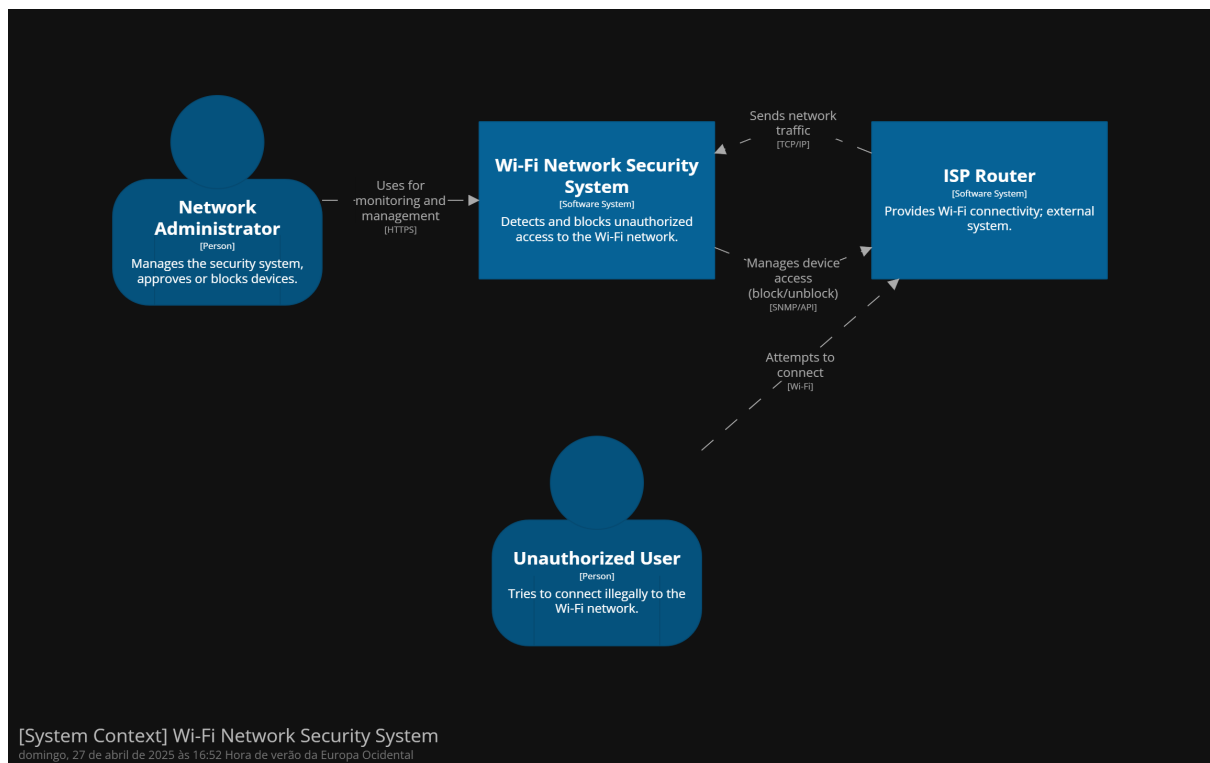


Figure 4: System Context Diagram

15.1.2 Level 2 : Container Diagram

The Container Diagram decomposes the system into major containers (i.e., applications or services) that run within the architecture. This view highlights how responsibilities are distributed across containers and how they communicate, particularly focusing on security and performance.

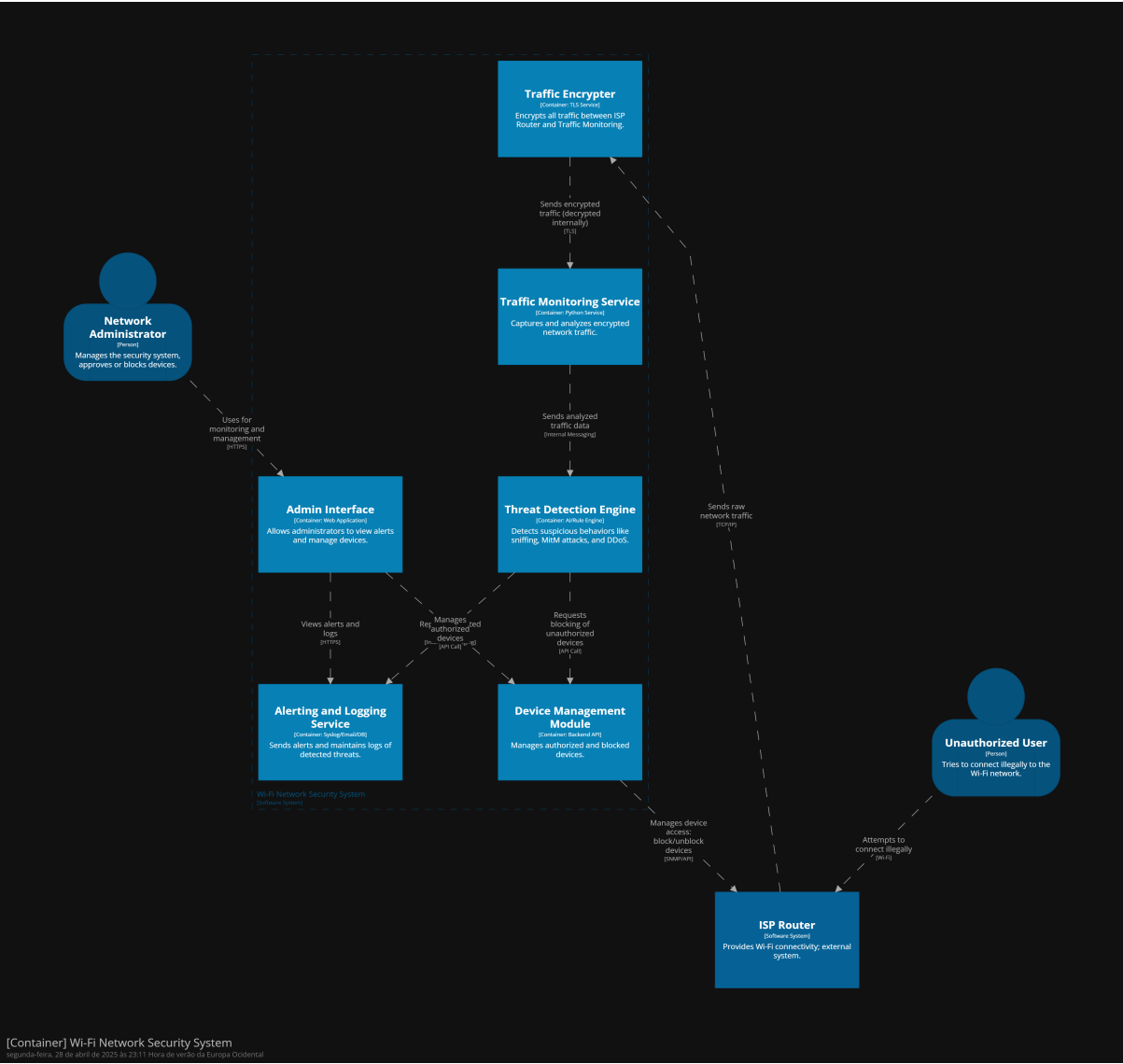


Figure 5: Wi-fi Network Security System Container Diagram

15.1.3 Level 3 : Component Diagram

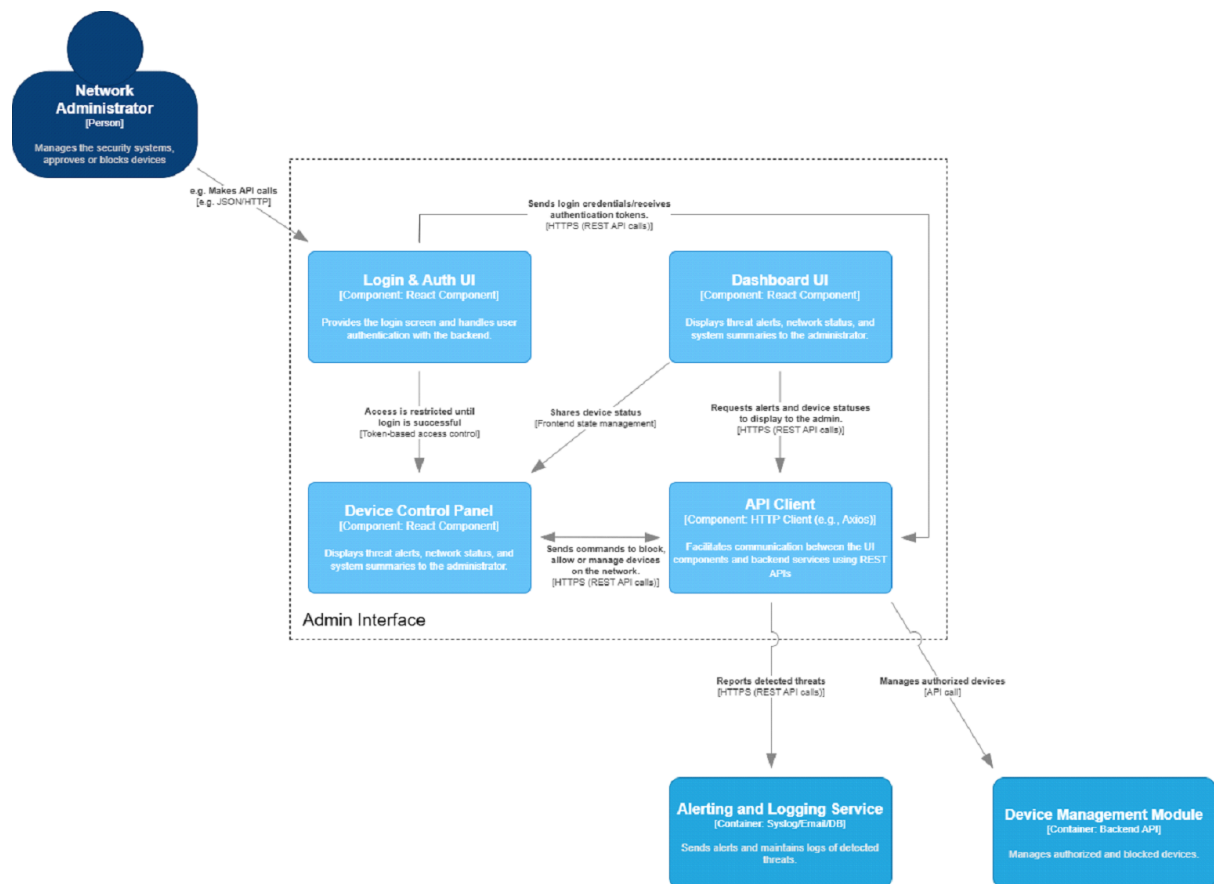


Figure 6: Component Admin Interface for Level 3 of C4 Model

16 Glossary

Glossary

DDoS Distributed Denial of Service — An attack that aims to make a system unavailable through excessive requests. 10

GDPR Regulamento Geral sobre a Proteção de Dados - A European legislation that aims to protect the personal data of citizens and businesses.. 7

HTTPS Hypertext Transfer Protocol Secure - It is the secure version of the HTTP protocol, which enables communication between a web browser and a web server.. 8

IoT Internet of Things - It is the network of physical objects connected to the internet, enabling data exchange and communication between them, such as sensors, software, and other technologies.. 9

ISP Internet Service Provider — entity that provides network access. 7

LAN Local Area Network - It is a network that connects computers and other devices within a relatively small area, such as a home.. 8

MAC address Mac Address - a unique identifier for a network interface controller (NIC) used to communicate on a network segment. 38

MFA Multi-Factor Authentication — Authentication with two or more factors (e.g.: password + SMS). 10

REST API An API architectural style that uses the HTTP protocol for data exchange. 8

TLS Transport Layer Security — Security protocol for encrypted communication. 17

VPN Virtual Private Network — Virtual private network used to ensure secure communications. 4, 31

Wi-Fi Wireless networking technology based on IEEE 802.11 standards. 4, 7